



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

TEMPERATURE INSTRUMENT PROCESS DATA SHEET

Docu. No. : 14780-8120-PS-0034
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

TEMPERATURE INSTRUMENT PROCESS DATA SHEET

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Concentration, Decomposition, and Neutralization Section

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TEMPERATURE INSTRUMENT

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Concentration, Decomposition, and Neutralization Section

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referenced in this	UOP Standard Drawing:
specification:	UOP Project Specification:



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TEMPERATURE INSTRUMENT

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Concentration, Decomposition, and Neutralization Section

ITEM NUMBER	SERVICE	SHEET
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Other specifications
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UOP Standard Specification:
UOP Standard Drawing:
UOP Project Specification:

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INSTRUMENT GENERAL REQUIREMENTS

PROJECT SPECIFICATION

 963766 — **601**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Phenol Concentration, Decomposition, and Neutralization Section

This specification shall be supplementary to all Project Specifications, 602 - 699, issued under this project number. Unless specified otherwise on the Project Specification, the following shall apply:

1 Climatic Ambient Temperature Range

The climatic ambient temperature range for field equipment design is 15 to 40°C.

2 Electrical Classification

The area electrical classification shall be established by the Licensee/Contractor in accordance with local electrical codes. The Contractor shall advise all equipment vendors of the area electrical classification.

3 Flanges

3.1 Carbon steel flanges are based on Material Group 1.1 of ASME B16.5 and ASME B16.47. Flange classes may differ when instrument flanges and the mating flanges on piping or equipment are specified with dissimilar materials. The Contractor shall verify that all mating flanges are properly matched with the specified instrument flanges.

3.2 Flange classes are specified in accordance with ASME B16.5 or ASME B16.47. Flange classes listed in the UOP Project Specifications are based upon design pressure and temperature conditions only and do not account for other loads. The final design of all flanges shall account for gasket seating and external loads. Differential thermal expansion, including joints between dissimilar materials, and transient thermal conditions such as start-up/shutdown and operational upset shall be accommodated.

3.3 Flanges intended for use with spiral wound gaskets shall have a flange surface finish of 125 microinch Ra minimum to 250 microinch Ra maximum. Flanges intended for use with other gaskets shall have a flange finish within the optimal range for the specified gasket. Finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1. Flange finishes shall be protected from damage during fabrication, heat treatment, shipping, storage, and installation.

4 Tropical Environment

All equipment shall be tropicalized to eliminate mildew, fungi, and other detrimental effects of a tropical environment. Packaging shall be suitable for shipment and storage under tropical conditions. Packaging and "tropicalization" subject to approval by the licensee.

5 Marine Environment

All field equipment shall be suitable for operation in a corrosive, salt laden, marine atmosphere.

6 Tropical and Marine Environment

Transmitters and all equipment/instruments containing electrical parts shall be installed under rain shield.

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INSTRUMENT GENERAL REQUIREMENTS

PROJECT SPECIFICATION

963766 — 601

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Phenol Concentration, Decomposition, and Neutralization Section

7 Fieldbus Transmitters

This specification is based on conventional (4 – 20 mA output) transmitters. UOP does not require the use of fieldbus transmitters and, if installed incorrectly, the fieldbus system can be detrimental to the control and operation of the unit. The implementation of these fieldbus transmitters is the Contractor's responsibility.

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THERMOCOUPLES AND WELLS

PROJECT SPECIFICATION

963766 — 606-G

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION

1 Standard Drawings Referenced

UOP Standard Drawings as indicated below are part of this specification.

X	6-105	X	6-107	X	6-108						
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2 Thermowell Length

For flanged thermowells the length from flange face to the tip of the thermowell (Length in UOP Standard Drawing 6-105) is based on the 6" nozzle projection indicated in UOP Standard Drawing 6-108. For insulated lines and vessels, a longer nozzle projection may be required, depending on the insulation thickness. The Contractor shall coordinate with equipment vendors and shall confirm the nozzle projection. For longer projections the thermowell length shall be increased as required to maintain the tip of the thermowell within one inch of the location determined from the dimensions given in this specification.

3 Assembly

When elements, terminal heads, connecting piping, thermocouple wells and protecting tubes are supplied by one vendor, the components shall be assembled into complete units. Upon assembly of the components, the element shall firmly contact the bottom of the well or tube.

4 Calibration Tolerances

Thermocouple initial calibration tolerances shall not be greater than the standard tolerances listed in ISA MC96.1-1982.

5 Substitution of Type E Thermocouples with Resistance Elements (RTD)

RTD Pt-100 (minimum of 3-wire) is an acceptable alternate to a type E thermocouple. The maximum operating temperature for RTD service shall be 850°F (450°C).

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing: See item 1 above
UOP Project Specification: 601, this project

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  				PROJECT SPECIFICATION 963766 — 606								
THERMOCOUPLES AND WELLS				REV	DATE	BY	APVD	REV	DATE	BY	APVD	
				0	25JUN12	MVD	MW					
Item Number: 1				Data Page: 1 of 1								
THERMOWELL	Type: UOP Standard Drawing 6-105, Detail A/B					ELEMENT	Type: E (Chromel-Constantan)					
	Connection Type: Flanged						Gauge: 5					
	Flange Facing: Raised Face Flange Face Finish:						Sheathed or Beaded Insulators:					
	Flange Material: 316 stainless steel						Sheath Outside Diameter:					
	Well Material: 316 stainless steel						Sheath Material:					
	Manufacturer: ThermoCouple Products (or equal)						Sheath Insulation Material:					
TERMINAL HEAD	Type: UOP Standard Drawing 6-107, Detail A						Junction: Grounded preferred					
	Manufacturer: ThermoCouple Products (or equal)						Manufacturer: ThermoCouple Products (or equal)					
INSTALLATION	Drawing: UOP Standard Drawing 6-108											
	Installation:											
REMARKS	§ The thermocouple wire gauge shall be 14 AWG or less for single or duplex element bare wire with beaded insulators and 18 AWG or less for sheathed single element and 19 AWG or less for sheathed duplex elements.											
	OP : Operating Pressure, DP : Design Pressure											
	OT : Operating Temperature, DT : Design Temperature											
DISTRIBUTION	Tag Number	Service	Well	Thermocouple	Terminal Head	Length	Size	Flange Class	Comments			
	TT/TE-23-0601	Concentration Cumene Quench Drum	1	1	1	600 mm	2"	300	OP : 0.06 kg/cm2(g) / OT : 38 degC			
									DP : 3.5 kg/cm2(g) / DT : 120 degC			
	TT/TE-23-1704	Reliable Cooling Water to Decomposer Cooler	1	1	1	300 mm	1-1/2"	150	OP : 4.71 kg/cm2(g) / OT : 37.4 degC			
									DP : 13.0 kg/cm2(g) / DT : 120 degC			
	TT/TE-23-1703	Reliable Cooling Water from Decomposer Cooler	1	1	1	300 mm	1-1/2"	150	OP : 2.5 kg/cm2(g) / OT : 43 degC			
DP : 13.0 kg/cm2(g) / DT : 120 degC												

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



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THERMOCOUPLES AND WELLS

PROJECT SPECIFICATION

963766 — 606

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

ITEM	Item Number: 2		Data Page: 1 of 3						
	Type: UOP Standard Drawing 6-105, Detail A/B Connection Type: Flanged Flange Facing: Raised Face Flange Face Finish: Flange Material: 304L stainless steel Well Material: 304L stainless steel Manufacturer: ThermoCouple Products (or equal)		Type: E (Chromel-Constantan) Gauge: 5 Sheathed or Beaded Insulators: Sheath Outside Diameter: Sheath Material: Sheath Insulation Material: Junction: Grounded preferred Manufacturer: ThermoCouple Products (or equal)						
TERMINAL HEAD	Type: UOP Standard Drawing 6-107, Detail A Manufacturer: ThermoCouple Products (or equal)								
INSTALLATION	Drawing: UOP Standard Drawing 6-108 Installation:								
REMARKS	§ The thermocouple wire gauge shall be 14 AWG or less for single or duplex element bare wire with beaded insulators and 18 AWG or less for sheathed single element and 19 AWG or less for sheathed duplex elements. OP : Operating Pressure, DP : Design Pressure OT : Operating Temperature, DT : Design Temperature								
DISTRIBUTION	Tag Number	Service	Well	Thermocouple	Terminal Head	Length	Size	Flange Class	Comments
	TT/TE-23-0301	Oxidate Feed to Preflash Column	1	1	1	300 mm	1-1/2"	150	OP : 5.5 kg/cm2(g) / OT : 82 degC DP : 11.5 kg/cm2(g), F.V / DT : 120 degC
	TT/TE-23-0402	Preflash Column Overhead Vapor	1	1	1	300 mm	1-1/2"	150	OP : -1.0 kg/cm2(g) / OT : 52 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TT/TE-23-0403	Preflash Column Bottoms	1	1	1	300 mm	1-1/2"	150	OP : -1.0 kg/cm2(g) / OT : 70 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TE-23-0404	Preflash Column Bottoms	1	1	1	300 mm	1-1/2"	150	OP : -1.0 kg/cm2(g) / OT : 70 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TT/TE-23-0506	Circulating Oxidate from Feed-Oxidate Exchangers	1	1	1	300 mm	1-1/2"	150	OP : 3.1 kg/cm2(g) / OT : 75 degC DP : 12.0 kg/cm2(g), F.V / DT : 120 degC
	TT/TE-23-0507	Circulating Oxidate from Feed-Oxidate Exchangers	1	1	1	300 mm	1-1/2"	150	OP : 3.1 kg/cm2(g) / OT : 75 degC DP : 12.0 kg/cm2(g), F.V / DT : 120 degC
	TE-23-0501	Preflash Column Feed-Oxidate Exchanger Tub Liquid	1	1	1	600 mm	2"	300	OP : -1.0 kg/cm2(g) / OT : 58 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TE-23-0502A	Preflash Column Steam Heater Tub Liquid	1	1	1	600 mm	2"	300	OP : -1.0 kg/cm2(g) / OT : 70 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.								

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PROJECT SPECIFICATION

963766 — 606

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

THERMOCOUPLES AND WELLS

Item Number: 2

REV	Tag Number	Service	Quantity			Length	Connection		Comments
			Well	Thermocouple	Terminal Head		Size	Flange Class	
DISTRIBUTION (continued)	TE-23-0502B	Preflash Column Steam Heater Tub Liquid	1	1	1	600 mm	2"	300	OP : -1.0 kg/cm2(g) / OT : 70 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TT/TE-23-0505	Preflash Column Steam Heater Tub Liquid	1	1	1	600 mm	2"	300	OP : -1.0 kg/cm2(g) / OT : 70 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TT/TE-23-0803	Flash Column Liquid Circulation to Flash Column Vaporizer	1	1	1	250 mm	1-1/2"	150	OP : -0.92 kg/cm2(g) / OT : 76 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TE-23-0805A	Flash Column Bottom	1	1	1	600 mm	2"	300	OP : -0.341 kg/cm2(g) / OT : 60 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TE-23-0805B	Flash Column Bottom	1	1	1	600 mm	2"	300	OP : -0.341 kg/cm2(g) / OT : 60 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TT/TE-23-0802	Flash Column Overhead Vapor	1	1	1	300 mm	1-1/2"	150	OP : -1.0 kg/cm2(g) / OT : 51 degC DP : 3.5 kg/cm2(g), F.V / DT : 250 degC
	TE/TT-23-0808	Preflash and Flash Columns Overhead Pumps Suction	1	1	1	300 mm	1-1/2"	150	OP : -0.4321 kg/cm2(g) / OT : 38 degC DP : 4.167 kg/cm2(g), F.V / DT : 250 degC
	TT/TE-23-0809	Oxidate Long Circulation to Oxidizer No.2 at Oxidation Section	1	1	1	300 mm	1-1/2"	150	OP : 2.36 kg/cm2(g) / OT : 38 degC DP : 13.55 kg/cm2(g) / DT : 250 degC
	TE-23-0901A	Flash Column Bottoms Pumps Suction	1	1	1	300 mm	1-1/2"	150	OP : -0.341 kg/cm2(g) / OT : 60 degC DP : 4.231 kg/cm2(g), F.V / DT : 195 degC
	TE-23-0901B	Flash Column Bottoms Pumps Suction	1	1	1	300 mm	1-1/2"	150	OP : -0.341 kg/cm2(g) / OT : 60 degC DP : 4.231 kg/cm2(g), F.V / DT : 195 degC
	TT/TE-23-1001	Flash Column Overhead Chiller Vent to Concentration Vacuum Producing Equipment	1	1	1	300 mm	1-1/2"	150	OP : -1.0 kg/cm2(g) / OT : 5 degC DP : 4.5 kg/cm2(g), F.V / DT : 120 degC
	TE-23-1301A	Decomposer Drum	1	1	1	600 mm	2"	300	OP : 0.7 kg/cm2(g) / OT : 60 degC DP : 11 kg/cm2(g), F.V / DT : 250 degC
	TE-23-1301B	Decomposer Drum	1	1	1	600 mm	2"	300	OP : 0.7 kg/cm2(g) / OT : 60 degC DP : 11 kg/cm2(g), F.V / DT : 250 degC
	TE-23-1302	Decomposer Drum	1	1	1	600 mm	2"	300	OP : 0.7 kg/cm2(g) / OT : 60 degC DP : 11 kg/cm2(g), F.V / DT : 250 degC

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PROJECT SPECIFICATION

963766 — 606

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

THERMOCOUPLES AND WELLS

Item Number: 2

DISTRIBUTION (continued)

Tag Number	Service	Quantity			Length	Connection		Comments
		Well	Thermocouple	Terminal Head		Size	Flange Class	
TT/ TE-23-1303	Decomposer Product Pumps Discharge	1	1	1	250 mm	1-1/2"	300	OP : 0.7 kg/cm2(g) / OT : 60 degC DP : 21 / FV kg/cm2(g) / DT : 250 degC
TE-23-1402	Decomposer Product from Dehydrators	1	1	1	250 mm	1-1/2"	300	OP : 4.9 kg/cm2(g) / OT : 140 degC DP : 21 / FV kg/cm2(g) / DT : 250 degC
TE-23-1401	Decomposer Product from Dehydrators	1	1	1	250 mm	1-1/2"	300	OP : 4.9 kg/cm2(g) / OT : 140 degC DP : 21 / FV kg/cm2(g) / DT : 250 degC
TT/ TE-23-1403	Decomposer Product to Crude Product Cooler	1	1	1	250 mm	1-1/2"	300	OP : 4.9 kg/cm2(g) / OT : 140 degC DP : 21 / FV kg/cm2(g) / DT : 250 degC
TE-23-1404	Crude Product from Crude Product Cooler	1	1	1	250 mm	1-1/2"	300	OP : 4.4 kg/cm2(g) / OT : 31 degC DP : 21 / FV kg/cm2(g) / DT : 250 degC
TT/ TE-23-1702 /1706	Circulating Decomposer Liquid from Decomposer Cooler	2	2	2	300 mm	1-1/2"	300	Fluid is at high velocity - additional well supports may be required. OP : 4.64 kg/cm2(g) / OT : 58 degC DP : 16.5 kg/cm2(g), F.V / DT : 250 degC
TT/ TE-23-1701 /1705	Circulating Decomposer Liquid to Decomposer Cooler	2	2	2	300 mm	1-1/2"	300	Fluid is at high velocity - additional well supports may be required. OP : 4.64 kg/cm2(g) / OT : 72 degC DP : 16.5 / FV kg/cm2(g) / DT : 250 degC
TT/ TE-23-1901	Crude Product to Direct Neutralization Static Mixer	1	1	1	250 mm	1-1/2"	300	OP : 4.4 kg/cm2(g) / OT : 31 degC DP : 21 / FV kg/cm2(g) / DT : 250 degC
TT/ TE-23-2001	Acid Aromatics Sump	1	1	1	mm	2"	300	Contractor to determine length such that thermowell tip is 150mm from sump bottom. OP : 0.01 kg/cm2(g) / OT : 38 degC DP : 3.5 / FV kg/cm2(g) / DT : 325 degC

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**PROJECT SPECIFICATION**963766 — **607-G****RESISTANCE ELEMENTS (RTD) AND WELLS**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION**1 Standard Drawings Referenced**

UOP Standard Drawings as indicated below are part of this specification.

X	6-105	X	6-107	X	6-108						
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2 Thermowell Length

For flanged thermowells the length from flange face to the tip of the thermowell (Length in UOP Standard Drawing 6-105) is based on the 6" nozzle projection indicated in UOP Standard Drawing 6-108. For insulated lines and vessels, a longer nozzle projection may be required, depending on the insulation thickness. The Contractor shall coordinate with equipment vendors and shall confirm the nozzle projection. For longer projections the thermowell length shall be increased as required to maintain the tip of the thermowell within one inch of the location determined from the dimensions given in this specification.

3 Assembly

When elements, terminal heads, connecting piping, resistance element wells and protecting tubes are supplied by one vendor, the components shall be assembled into complete units. Upon assembly of the components, the element shall firmly contact the bottom of the well or tube.

4 Operating Temperature Limit

The maximum operating temperature for RTD service shall be 850°F (450°C).

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing: See item 1 above
UOP Project Specification: 601, this project



PROJECT SPECIFICATION
963766 — 607

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

RESISTANCE ELEMENTS (RTD) AND WELLS

ITEM NO.	Item Number: 1		Data Page: 1 of 1						
	THERMOWELL Type: UOP Standard Drawing 6-105, Detail A/B Connection Type: Flanged Flange Facing: Raised Face Flange Face Finish: Flange Material: 304L stainless steel Well Material: 304L stainless steel Manufacturer: ThermoCouple Products (or equal)		ELEMENT	Type: Platinum 100 ohm @ 0°C Style: Number of Leads: Minimum of 3 Wire Size: Manufacturer's standard Sheath Outside Diameter: Manufacturer's standard Sheath Material: 304L/316 stainless steel Sheath Insulation Material: Magnesium oxide Manufacturer: ThermoCouple Products (or equal)					
TERMINAL HEAD	Type: UOP Standard Drawing 6-107, Detail F Manufacturer: ThermoCouple Products (or equal)								
INSTALLATION	Drawing: UOP Standard Drawing 6-108 Installation:								
REMARKS	OP : Operating Pressure, DP : Design Pressure OT : Operating Temperature, DT : Design Temperature								
DISTRIBUTION	Tag Number	Service	Well	RTD Element	Terminal Head	Length	Size	Flange Class	Comments
	TE-23-0702	Flash Column Vaporizer Outlet	1	1	1	300 mm	2"	300	OP : -1.0 kg/cm2(g) / OT : 97 degC DP : 4.0 kg/cm2(g), F.V / DT : 250 degC
	TE-23-0703	Flash Column Vaporizer Outlet	1	1	1	300 mm	2"	300	OP : -1.0 kg/cm2(g) / OT : 97 degC DP : 4.0 kg/cm2(g), F.V / DT : 250 degC
	TE-23-0701	Flash Column Vaporizer	1	1	1	300 mm	2"	300	OP : -1.0 kg/cm2(g) / OT : 97 degC DP : 4.0 kg/cm2(g), F.V / DT : 250 degC

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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 A Honeywell Company    RESISTANCE ELEMENTS (RTD) AND WELLS				PROJECT SPECIFICATION 963766 — 607 <table><tr><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td></tr><tr><td>0</td><td>25JUN12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	25JUN12	MVD	MW																												
REV	DATE	BY	APVD	REV	DATE	BY	APVD																																												
0	25JUN12	MVD	MW																																																
Item Number: 2				Data Page:1 of 2																																															
THERMOWELL	Type:				ELEMENT	Type: Platinum 100 ohm @ 0°C																																													
	Connection Type:					Style:																																													
	Flange Facing:	Flange Face Finish:				Number of Leads: Minimum of 3																																													
	Flange Material:					Wire Size: Manufacturer's standard																																													
	Well Material:					Sheath Outside Diameter: Manufacturer's standard																																													
	Manufacturer:					Sheath Material: 304/316 stainless steel																																													
	Type: Sketch A			Sheath Insulation Material: Magnesium oxide																																															
TERMINAL HEAD	Manufacturer: ThermoCouple Products (or equal)			Manufacturer: ThermoCouple Products (or equal)																																															
INSTALLATION	Drawing: Sketch B																																																		
	Installation: Duplex RTD element shall be installed through a 1/4" compression fitting such that the tip is located in the center of a 1/2" class 3000 socket weld tee assembly. See Sketch B.																																																		
REMARKS	The contractor shall arrange the critical volume piping layout to permit installation of the tee assembly.																																																		
	OP : Operating Pressure, DP : Design Pressure																																																		
	OT : Operating Temepreature, DT : Design Temperature																																																		
DISTRIBUTION	Tag Number	Service	Well	RTD Element	Terminal Head	Length	Size	Flange Class	Comments																																										
	TT/TE-23-1807	Circulating Decomposer Liquid from Calorimeter No. 1	~	1	1	150 mm	~	~	OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC																																										
	TE-23-1806 B,1801B	Circulating Decomposer Liquid from Calorimeter No. 1	~	2†	1	150 mm	~	~	† Duplex OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC																																										
	TE-23-1806 A,1802B	Circulating Decomposer Liquid to Calorimeter No. 1 Inlet	~	2†	1	150 mm	~	~	† Duplex OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC																																										
	TT/TE-23-1805	Circulating Decomposer Liquid to Calorimeter No. 1 Inlet	~	1	1	150 mm	~	~	OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC																																										
	TE-23-1802 A,1801A	Circulating Decomposer Liquid to Calorimeter No. 1 Line Beginning Inlet	~	2†	1	150 mm	~	~	† Duplex OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 16.5 kg/cm2(g), F.V / DT : 250 degC																																										
	TE-23-1804 A,1803A	Circulating Decomposer Liquid to Calorimeter No. 2 Line Beginning Inlet	~	2†	1	150 mm	~	~	† Duplex OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 16.5 kg/cm2(g), F.V / DT : 250 degC																																										
	TT/TE-23-1808	Circulating Decomposer Liquid to Calorimeter No. 2 Inlet	~	1	1	150 mm	~	~	OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC																																										
	TE-23-1804 B,1809A	Circulating Decomposer Liquid to Calorimeter No. 2 Inlet	~	2†	1	150 mm	~	~	† Duplex OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC																																										
	ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																		





RESISTANCE ELEMENTS (RTD) AND WELLS

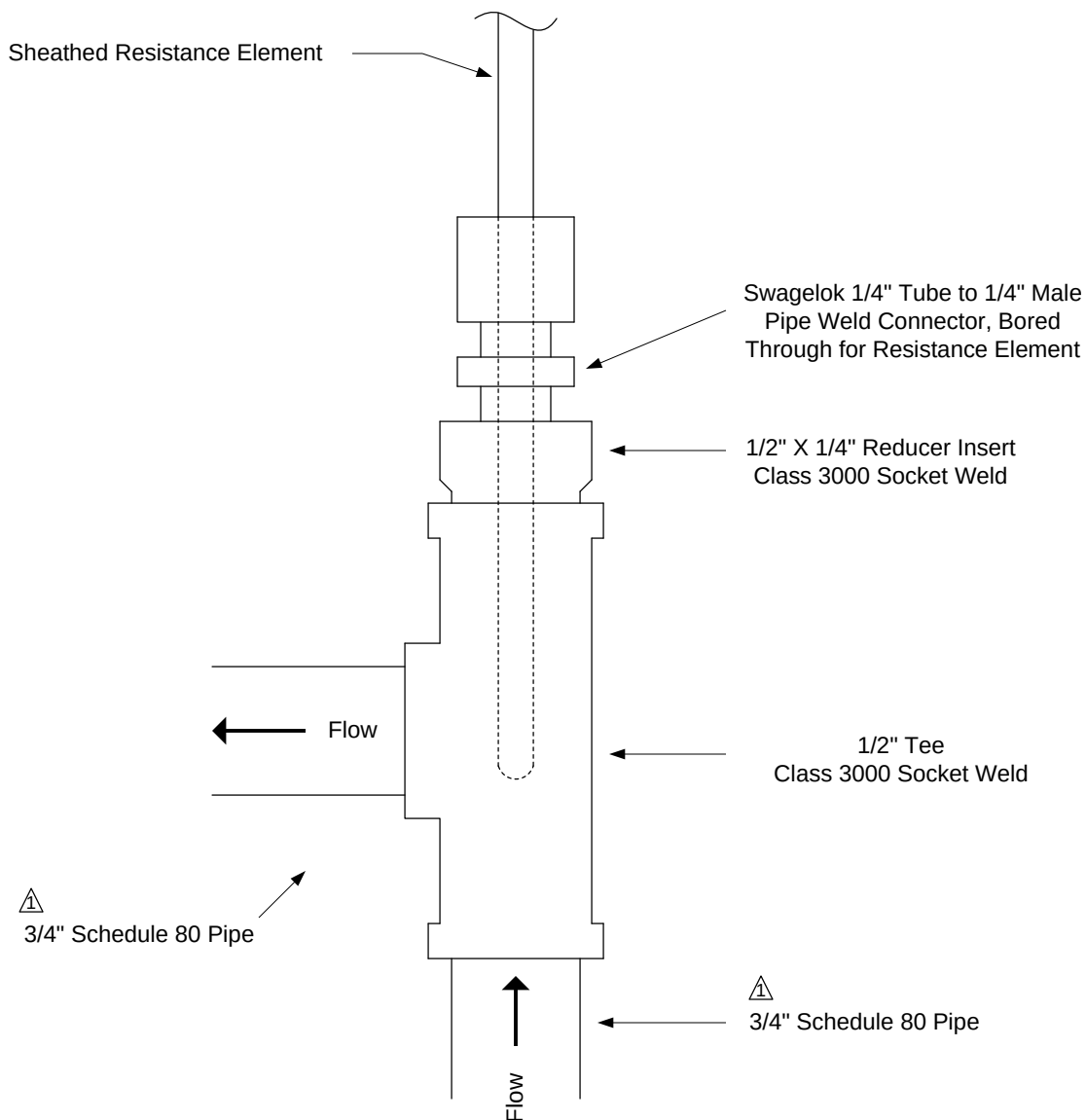
PROJECT SPECIFICATION							
963766 —				607			
REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

	REV	Item Number: 2								
		Tag Number	Service	Quantity			Length	Connection		Comments
				Well	RTD Element	Terminal Head		Size	Flange Class	
DISTRIBUTION (continued)		TE-23-1809 B,1803B	Circulating Decomposer Liquid from Calorimeter No. 2	~	2†	1	150 mm	~	~	† Duplex OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC
		TT/ TE-23-1810	Circulating Decomposer Liquid from Calorimeter No. 2	~	1	1	150 mm	~	~	OP : 4.46 kg/cm2(g) / OT : 79 degC DP : 10.5 kg/cm2(g), F.V / DT : 250 degC

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**PROJECT SPECIFICATION**963766 — **607-D****RESISTANCE ELEMENTS (RTD) AND WELLS**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Sketch " B "**Notes:**

1. Swagelok tube to pipe weld connector shall be provided with stainless steel ferrule compression seal.
2. All components shown shall be 304 stainless steel.

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TEMPERATURE INSTRUMENTS (FIELD)

PROJECT SPECIFICATION

963766 — 609

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

General Requirements and Information

1. Inputs

- a. Loop powered transmitters shall have the input isolated from the output. Transmitters with internal power supplies shall have the input isolated from the output and power supply.
- b. RTD: 3 or 4-wire Pt 100
- c. Thermocouples: Type E, J, K, and T and differential thermocouple
Thermocouple transmitters shall have automatic cold junction compensation.

2. Output

- a. Transmitters shall be calibrated for the measurement range specified in the UOP Project Specifications.
- b. The transmitter output for the specified range can be 4 – 20 mA, or 4 – 20 mA with a digital communication protocol superimposed, or a fully digital communication protocol. The communication protocol shall not prevent the instrument from measuring and transmitting the process temperature, except when the instrument is being configured.
- c. For transmitters with 4 – 20 mA output, self-diagnostics shall be able to detect both a thermocouple/RTD sensor failure and transmitter electronics failure. The transmitter shall be user selectable to drive the output upscale or downscale to a predetermined value upon failure detection (such as NAMUR NE 43 engl. compliant transmitters). Upscale drive on failure detection shall be selected unless specified otherwise.
- d. Fieldbus transmitters shall be capable of detecting and transmitting sensor failure or electronics failure. The fieldbus host shall be capable of detecting when a transmitter has been added or removed from the bus and when the host has lost communication with the transmitter.
- e. Transmitters in control services shall have an update frequency of at least two times per second. Both response time and update rate shall be considered prior to the selection of the equipment

3. Performance

- a. Electronic analog transmitter accuracy shall be $\pm 0.5\%$, or better, of calibrated span and includes the effects of repeatability, hysteresis, and linearity.
- b. Digital, micro-processor based
 1. Digital accuracy $< \pm 0.5^\circ\text{F}$ (0.28°C) for RTDs and $< \pm 1.0^\circ\text{F}$ (0.55°C) for thermocouples.
 2. Digital to Analog conversion accuracy for milliAmp outputs shall be less than $\pm 0.05\%$ of calibrated span.
 3. Combined digital accuracy for a 50°F (28°C) change in ambient temperature $< \pm 1.0^\circ\text{F}$ (0.55°C) for RTDs and $< \pm 2.0^\circ\text{F}$ (1.1°C) for thermocouples.
 4. Transmitter output shall be linear with temperature.

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TEMPERATURE INSTRUMENTS (FIELD)

PROJECT SPECIFICATION

963766 — 609

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

4. Transmitter Location

Temperature transmitters may be located either in the field, control room, or field cabling room as long as the ambient temperature limits for the transmitter are not exceeded. For locations with large ambient temperature changes, either daily or seasonally, it is recommended that the transmitters be located in a temperature controlled environment to limit measurement errors due to ambient changes. When transmitters are located near hot process lines or vessels, provide sufficient thermowell/sensor extension to protect the transmitters from excessive temperatures.

5. Analog to Digital Conversion

In control services, an analog-to-digital conversion of the direct thermocouple millivolt or RTD signal over the entire thermocouple/RTD sensor range in the digital control system may be used instead of transmitters if the temperature resolution is 0.05°F (0.028°C) or less.

6. Miscellaneous Plant Temperature Indicators

Miscellaneous plant temperature indicator points are those points that do not require a narrow temperature indication range. When field mounted temperature transmitters are used for such points, then for those transmitters not specified by tag number in this specification, the following table may be used for selecting the transmitter ranges.

The following criteria may be used for selecting the Measurement Ranges for Display:

- 30 to 200°C for maximum operating temperature ≤ 150°C
- 30 to 400°C for maximum operating temperature > 150°C and ≤ 350°C
- 30 to 650°C for maximum operating temperature > 350°C and ≤ 600°C
- 30 to 850°C for maximum operating temperature > 600°C and ≤ 800°C
- 30 to 1100°C for maximum operating temperature > 800°C

For cold service applications, the lower range value shall be approximately 30°C below the lowest expected operating temperature, and the upper range value shall be +50°C.

- 20 to 400°F for maximum operating temperature ≤ 300°F
- 20 to 750°F for maximum operating temperature > 300°F and ≤ 650°F
- 20 to 1200°F for maximum operating temperature > 650°F and ≤ 1100°F
- 20 to 1600°F for maximum operating temperature > 1100°F and ≤ 1500°F
- 20 to 2000°F for maximum operating temperature > 1500°F

For cold service applications, the lower range value shall be approximately 50°F below the lowest expected operating temperature, and the upper range value shall be +120°F.

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TEMPERATURE INSTRUMENTS (FIELD)

PROJECT SPECIFICATION

963766 — 609

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description
------	----------	-------------

1	as required	Provide the following temperature transmitters for field installation:
---	-------------	--

Supplier (or equal):

Rosemount Inc.
8200 Market Blvd.
Chanhassen, Minnesota 55317
USA
www.emersonprocess.com

Foxboro (Invensys Systems, Inc.)
33 Commercial Street
Foxboro, Massachusetts 02035
USA
www.foxboro.com

Tag Number	Service	Type	Calibrated Range (°C)	Remarks
TXT -23-0404	Preflash Column Bottoms	E	0 - 200	
TXT -23-0501	Preflash Column Feed-Oxidate Exchanger Tub Liquid	E	0 - 200	
TXT -23-0502A	Preflash Column Steam Heater Tub Liquid	E	0 - 200	
TXT -23-0502B	Preflash Column Steam Heater Tub Liquid	E	0 - 200	
TXT -23-0701	Flash Column Vaporizer Outlet Elbow	RTD	0 - 200	
TXT -23-0702	Flash Column Vaporizer Outlet	RTD	0 - 200	
TT-23-0703	Flash Column Vaporizer Outlet	RTD	60 - 120	
TXT -23-0805A	Flash Column Bottom	E	0 - 200	
TXT -23-0805B	Flash Column Bottom	E	0 - 200	
TXT -23-0901A	Flash Column Bottoms Pumps Suction	E	0 - 200	
TXT -23-0901B	Flash Column Bottoms Pumps Suction	E	0 - 200	
TXT -23-1301A	Decomposer Drum	E	0 - 200	
TXT -23-1301B	Decomposer Drum	E	0 - 200	
TT-23-1302	Decomposer Drum	E	40 - 90	
TXT -23-1402	Decomposer Product from Dehydrators	E	0 - 200	
TT-23-1401	Decomposer Product from Dehydrators	E	0 - 200	
TT-23-1404	Crude Product from Crude Product Cooler	E	0 - 100	TI-23-1404B must be readable from valve.
TDXT -23-1801	Calorimeter No. 1 Total System Differential Temperature	RTD	0 - 40	Differential Temperature

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing:
UOP Project Specification: 601, this project

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TEMPERATURE INSTRUMENTS (FIELD)

PROJECT SPECIFICATION

963766 — 609

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item

1 (cont'd)

Tag Number	Service	Type	Calibrated Range (°C)	Remarks
TDT-23-1806	Calorimeter No. 1 Differential Temperature	RTD	0 - 25	Differential Temperature
TDXT -23-1802	Calorimeter No. 1 Inlet Line Differential Temperature	RTD	0 - 20	Differential Temperature
TDXT -23-1804	Calorimeter No. 2 Inlet Line Differential Temperature	RTD	0 - 20	Differential Temperature
TDT-23-1809	Calorimeter No. 2 Differential Temperature	RTD	0 - 25	Differential Temperature
TDXT -23-1803	Calorimeter No. 2 Total System Differential Temperature	RTD	0 - 40	Differential Temperature

Revision
Indication

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SPECIAL TEMPERATURE DETECTION INSTRUMENTS

PROJECT SPECIFICATION

963766 — 640

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Item	Quantity	Description
1	1	Flanged thermowell assembly including a resistance element (RTD), flexible conduit, accessories, and terminal head for terminating the element. Tag Number and Service: TE-23-0701 Flash Column Vaporizer Tube Thermowell Assembly Thermowell flange to be 2" Class 300 RF 304L SS. Thermowell material shall be seamless 4.7625 mm (3/16") O.D., 0.712 mm (0.028") wall thickness 304L SS tubing. Thermowell to be fabricated from a single piece of tubing. See Sketch "A". Thermowell shall be welded to bottom center of the flange with ½" NPT threaded fitting screwed into the flange and seal welded to the thermowell. A 1/8" half coupling shall be welded to the top of the flange and a 1/8" NPT, 1/8" Thermocouple Products (or equal) thermocouple gland will be screwed into the half coupling. Gland shall be furnished with lava seal. The contractor shall determine the thermowell length required based on the exchanger purchased, to position the thermocouple tip inside an exchanger tube, 24 inches (600 mm) below the top of the tube bundle. See Sketch "A". Three radial guide wires oriented 120 degrees apart shall be welded to the thermowell approximately 6 inches (150 mm) above the thermowell tip, to keep the thermowell centered inside the heat exchanger tube. The radial height of the wires shall suit the heat exchanger tube internal diameter, to allow the thermowell to fit easily inside the tube, with a nominal 0.2 mm clearance between the end of the guide wires and the tube wall when the thermowell is centered in the tube. The contractor shall determine the heat exchanger tube inside diameter. Furnish one 16 AWG resistance element (RTD) in a mineral oxide insulated, 1/8" annealed 304 stainless steel sheath. The sheathed element shall extend approximately 60 mm above the thermocouple gland, and shall be terminated with a stainless steel moisture and weatherproof transition fitting. Connector bodies shall be 18-8 stainless steel. Resistance element and extension wire connections shall be silver soldered or welded. See Sketch "B". Process Conditions: Temperature: 97 °C Pressure: Full Vacuum Design Conditions: 4.0 kg/cm2(g) at 250 °C Supplier (or equal): Thermo-Couple Products Co. State Route 2 Box 305 Newell, West Virginia 26050 USA www.tcproducts.com

 Other specifications
referenced in this
specification:

 UOP Standard Specification:
 UOP Standard Drawing:
 UOP Project Specification: 601, this project

Revision
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SPECIAL TEMPERATURE DETECTION INSTRUMENTS

PROJECT SPECIFICATION

963766 — 640

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

1 (cont'd)

Notes

1. The thermowell assembly must be pressure tested for 170 psig (12.0 kg/cm2(g)) minimum.
2. Resistance element shall be tested prior to shipping.
3. The element shall be removed from the thermowell after initial assembly and then reinstalled to assure proper fitting connections.
4. The supplier shall furnish a complete spare parts list including resistance elements.
5. The supplier shall furnish a calibration curve for the resistance element in the range of 40-140 °C.
6. Provide a minimum of three spare lava seals for the assembly.

Revision
Indication**PROJECT SPECIFICATION**

963766 — 640

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

**SPECIAL TEMPERATURE
DETECTION INSTRUMENTS****Sketch " A "**

Long Pipe Thermowell for use with Mineral Oxide Insulated Metal Sheathed
Resistance Element (RTD)

1/8" NPT x 0.125 Mounting Adapter
(304 Stainless Steel, See Note)

1/8" Half Coupling

2" ASME B16.5 Flange,
Class 300RF

Seal Weld

1/2" NPT threaded fitting with inside
diameter to match thermowell
outside diameter

100 mm (nominal;
vendor to confirm)

3/16" O.D.
22 BWG Tubing

Length

4" (100 mm)

Stitch Weld at
both ends
(typical)

6" (150 mm)

120°

To suit heat
exchanger tube
I.D.

Wire gauge shall be
approximately 0.040"
to 0.062" in
diameter

Note:

The mounting adapter shall be
provided with a lava seal.
Provide three spare lava seals

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SPECIAL TEMPERATURE DETECTION INSTRUMENTS



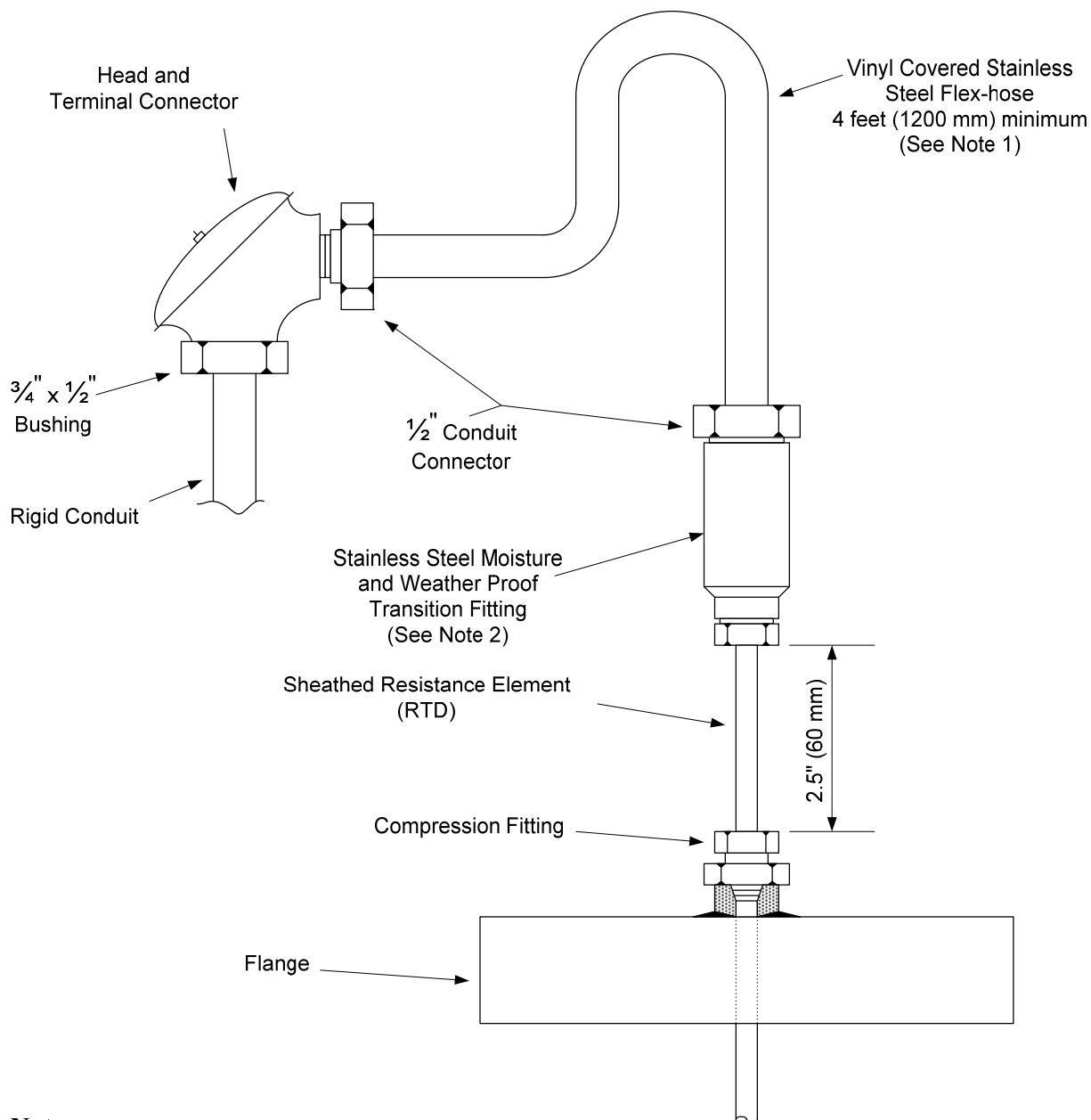
PROJECT SPECIFICATION

963766 — 640

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Sketch " B "

Resistance Element (RTD) Terminal Head Assembly for use with Mineral Oxide Insulated Metal Sheathed Resistance Element (RTD)



Notes:

1. The supplier of the assembly shall provide 16 AWG insulated extension wires between the transition fitting and the terminal connector.
2. The wires leaving the sheath shall be silver soldered to 16 AWG resistance element (RTD) extension wire within the transition fitting.



TEMPERATURE INSTRUMENT PROCESS SPECIFICATION

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117			
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300			
REV		O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0034		
DATE		DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016			
PREP. BY		S.W.Park	S.W.Park	S.W.Park	S.W.Park			
REVIEW BY		J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung			
APP. BY		S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee			
	1	General	Item No.	TI-23-0807		TI-23-0808	TI-23-1002	TI-23-1405
			P&ID No.	14780-8120-25-23-0008		14780-8120-25-23-0008A	14780-8120-25-23-0010	14780-8120-25-23-0014A
			Service	Reliable Cooling Water from Flash Column Bottoms Cooler		HP Reliable Cooling Water to Preflash and Flash Columns Condenser	Chilled Water from Flash Column Overhead Vapor Chiller	Cooling Water from Crude Product Cooler
	3	Line Size inch		6"	28"	6"	10"	
	4	Fluid Properties	Type					
			State	LIQ	LIQ	LIQ	LIQ	
			Description	RELIABLE COOLING WATER	RELIABLE COOLING WATER	CHILLED WATER	COOLING WATER	
			Corrosion/ Erosion due to					
	5							
	21	Operating temp °C	Minimum					
			Normal	43	33	5	43	
			Maximum					
	22	Operating Pressure kg/cm ² g Normal		2.5	6.4	2.5	2.5	
	25	Velocity m/s						
	32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	
	33	High Alarm		N/A	N/A	N/A	N/A	
	34	Low Alarm		N/A	N/A	N/A	N/A	
	35	Design Pressure kg/cm ² (G)		10.5	10.5	11.5	7	
	36	Design Temperature °C		120	120	120	120	
	37	Remarks						
	38	Revision						

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



TEMPERATURE INSTRUMENT PROCESS SPECIFICATION

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117	
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300	
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0034	
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016		
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park		
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung		
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee		
1	General	Item No.	-	TI-23-0704		
		P&ID No.	-	14780-8120-25-23-0001E		
			-	SN-2310		
		Service				
3	Line Size	inch	-	4		
4	Fluid Properties	Type	-	3/4		
		State	-	LIQ		
		Description	-	HYDROCARBON		
		Corrosion/ Erosion due to	-			
5			-			
21	Operating temp °C	Minimum	-			
		Normal	-	70		
		Maximum	-			
22	Operating Pressure	kg/cm ² g	Normal	-	-1	
25	Velocity	m/s	-			
32	Instrument Range	(From/ To)	-	(NOTE 1)		
33	High Alarm		-	N/A		
34	Low Alarm		-	N/A		
35	Design Pressure	kg/cm ² (G)	-	4 / FV		
36	Design Temperature	°C	-	250		
37	Remarks		DELETED			
38	Revision					

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).

Revision
Indication



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 A Honeywell Company		UOP LLC 25 E. Algonquin Road Des Plaines, Illinois 60017-5017, USA		PROJECT SPECIFICATION 963766 — 696					
FIELD OUTPUT INDICATORS		REV	DATE	BY	APVD	REV	DATE	BY	APVD
		0	25JUN12	MVD	MW				
Item	Quantity	Description							
1	as required	Electronic output indicator actuated by a 4 - 20 mA signal (Contractor to verify) from a field transmitter or from the control center.							
		Tag Number		Service				Scale	
		TI-23-1404B		Crude Product from Crude Product Cooler				0 - 100 °C	
Supplier (or equal): Rosemount www.emersonprocess.com									
Other specifications referenced in this specification:		UOP Standard Specification:							
		UOP Standard Drawing:							
		UOP Project Specification: 601, this project							